

Homework — 2.3.1 Algorithms

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 — set after the 2.3.1 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. A search algorithm halves the portion of a sorted list it is examining on every step.

(a) State its Big O time complexity. **[1]** (AO1)

(b) **Justify** this complexity in one sentence, and state the **precondition** the list must satisfy. **[1]** (AO2)

2. Trace a **bubble sort** (one pass = compare adjacent pairs left-to-right, swapping if out of order) on the list below. Show the list **after each complete pass** until it is sorted. **[3]** (AO2)

Start: [6, 2, 9, 4, 1]

After pass	Item 1	Item 2	Item 3	Item 4	Item 5
Pass 1					
Pass 2					
Pass 3					
Pass 4					

3. A **binary search** is performed on the sorted list below to find the value **31**.

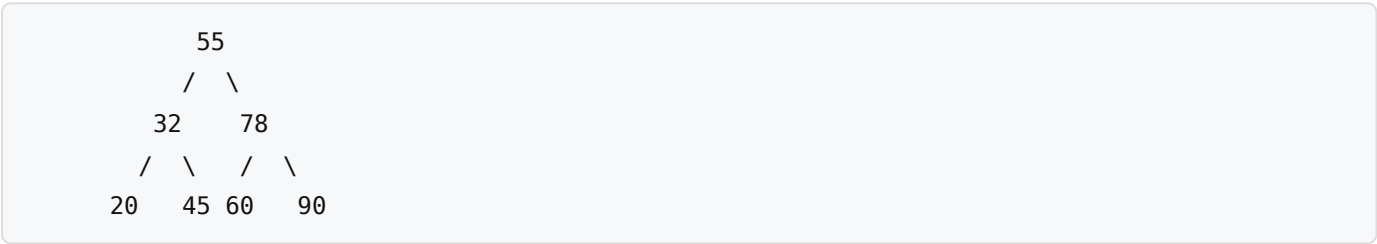
[4, 9, 15, 18, 23, 27, 31, 42, 56] (indices 0–8)

Complete the trace table (use integer division for mid). Show each step until the value is found. **[3]** (AO2)

Step	low	high	mid (index)	value at mid	< = > target?
1					
2					

State the result: _____

4. Consider the binary search tree below.



Write the output of each traversal below. (AO2)

(a) In-order: _____ [1]

(b) Pre-order: _____ [1]

(c) Post-order: _____ [1]

5. Dijkstra's algorithm is run on the weighted graph below, starting at node A, to find the shortest distance to every other node.

Edges (undirected, with weights): A–B = 2, A–C = 5, B–C = 1, B–D = 7, C–D = 3, C–E = 8, D–E = 2

Complete the table of final shortest distances from A. [3] (AO2)

Node	Shortest distance from A	Via (previous node)
A	0	—
B		
C		
D		
E		

Part B — Retrieval: keep it fresh (~6 marks)

6. (2.3.1) State whether **merge sort** or **quick sort** guarantees $O(n \log n)$ in the worst case, and give the **worst-case** complexity of the other. [2]

7. (2.3.1 / 1.4.2) State the data structure used by a **breadth-first traversal**, and explain why that structure is appropriate. [2]

8. (2.2.2) A problem can only be solved in **exponential time or worse**. State the term used to describe such a problem, and give **one** example. [2]

Part C — Exam-style question (~5 marks)

9. A logistics company needs to calculate the shortest delivery distance from a single depot to **every** delivery point on a road network where all distances are positive.

Recommend either **Dijkstra's algorithm** or the **A*** algorithm for this task, and **justify** your choice by comparing the two. **[5]** (AO3)

[illegible]